general motors

CG4209 GM 1927 03a
SQ - Electrical & Electronic Modules & Assemblies

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1. Introduction

Note: Nothing in this standard shall supersede applicable laws and regulations. In the event of a conflict between English and a domestic language, the English language shall take precedence.

1.1. **Purpose**. The following minimum requirements are to be incorporated into the manufacturing processes for Electrical & Electronic Modules and Assemblies. It is the responsibility of the supplier to ensure that the manufacturing process is state of the art.

1.2. **Applicability.** These requirements are in addition to any requirements as outlined in the CG4338 GM 1927 03 Supplier Quality SOR. “Shall” in this document is mandatory. “Should” is highly recommended.

1.2.1. These requirements are valid for any components or assemblies manufactured at the sub-supplier (Tier 1 or Tier 2… Tier x.).

1.3. **Remarks.** The required tasks indicated below are based on lessons learned to improve part quality using APQP Continuous Improvement activity in GM projects and are applicable to all impacted suppliers and parts in the supply chain.

1.3.1. All deviations requested for “shall” items are to be detailed in CG3404 M7 Technical Issues List found in eSOR Appendix M7 for sourcing or captured in the APQP issues action plan and reviewed by the GM Supplier Quality SMT Manager.

1.3.2. The supplier shall use this document (Electrical & Electronic Modules & Assemblies) as an audit checklist. It shall be completed by the time of the APQP Kickoff review and verified at each applicable CRV

2. References

Note: Only the latest approved standards are applicable unless otherwise specified.

External Standards/Specifications [L]

Semiconductor / Fabless Manufactures/ BKD & Assembly		**Tier 1 / Contract Manufacturers / PCB suppliers**	
ANSI/ESD S20.20 or IEC 61340-5.1	ANSI/ESD SP27.1-2018	ANSI/ESD S20.20 or IEC 61340-5.1	ANSI/ESD SP27.1-2018
AEC-Q100	AEC Q001	JEDEC 243	IPC-DR-572A
AEC-Q101	AEC Q002	AIAG: CQI-17	IPC 6010
AEC-Q102	AEC Q003	IPC-A-610	IPC 7527
AEC-Q103	AEC Q004	IPC 7711/7721	IPC 5704
AEC-Q200	AEC Q005	IPC 5703	IPC 1601
J-STD-046 or ZVEI	AEC Q006	IPC-6012	J-STD-004B
J-STD-048 or ZVEI		IEC 61340-5-X	IEC-60352-5
JEDEC J-STD-033		JEDEC J-STD-033	JEDEC J-STD-020

2.1. GM Standards/Specifications [D], [E], [F], [H]

CG2503	GM3660	SQ SOR, Part Specific CGs	Process Specific Audits
CG2999	GMW17407	(CG4338 GM 1927 03, 03a)	(GM 1927 16b)
CG3001			

3. Requirements

3.1. An assessment using AIAG CQI – 17 *Soldering System Assessment Requirements* shall be conducted and reviewed at the tech review. Open issues shall be captured on the APQP Open Issues List and closed prior to PV parts fabrication.

3.1.1. All findings classified as "Fail", or "Needs Immediate Action", or "Not Satisfactory" should be included in a continuous improvement plan and could, at the discretion of the SQE, impact Full PPAP status.

3.1.2. The "*Special Process: Soldering System Assessment-General Facility Overview"* results should be conducted annually, and the results summarized.

3.1.2.1. For any non-compliances, a regular audit using the "GM 1927 16b Electronic Assembly Manufacturing-Soldering Process Audit" should be used to control areas of risk noted in the General Facility Overview.

3.2. The Supplier manufacturing process shall incorporate the acceptance requirements of electrical assemblies as described in the latest version of IPC-A-610 Class 3 standard.

3.2.1. Wave-soldering shall be avoided for new products/New SORs. If used, the soldering area shall be encased in nitrogen for process stability, or long-term process capability data shall be provided to show stability.

3.2.2. Soldering by hand shall not be used in normal production [B]. It may be used in rework areas with special trained personnel for authorized components. Any approved rework process shall be validated and included in the work instructions and process control plan.

3.3. The Supplier manufacturing process shall incorporate the acceptance requirements of electrical assemblies as described in the IPC-7721/7711 rework / repair standard as well as the following additional requirements.

3.3.1. If reworks are allowed for a module/assembly, they shall be inspected by someone different from who performed the rework per IATF 16949 8.7.1.4 Control of reworked product and 8.7.1.5 Control of repaired product requirements.

3.3.2. Rework / solder touch up shall be avoided. With documented approval of the validated and stabilized process, certain types (e.g. cost, complexity) of rework / solder touch up will be allowed.

3.3.2.1. Analysis of the failures shall be done to identify potential process improvements.

3.3.2.2. A maximum of 3 different components on a PCB and a maximum 1 rework attempts on a specific component location are allowed. The goal should be 1 or less.

3.3.2.3. Reworked / solder touched up parts are required to re-enter the line in the appropriate station to guarantee that all process steps (inspection, programming, testing, etc.) have been fully completed.

3.3.3. The following coating removal/replacement procedures are not allowed:

3.3.3.1. IPC-7721/7711 item 2.3.5: Grinding Scraping Method

3.3.3.2. IPC-7721/7711 item 2.3.6: Micro Blasting Method

3.3.3.3. IPC-7721/7711 item 2.4.1: Solder Resist (because Part 3 shall not be used).

3.3.3.4. Any flux removal method that leaves a flux residue that promotes dendritic growth.

3.3.4. The following rework procedures are not allowed:

3.3.4.1. IPC-7721/7711 5.7.1(Using wire Solder to Pre-fill Lands) shall not be used for component reworks.

3.3.4.2. IPC-7721/7711 8.1(wiring rework: splicing) shall not be used. The wire shall be replaced with a new one.

3.3.4.3. IPC-7721/7711 Part 3 shall not be used.

3.3.5. Reworked parts and assemblies shall have traceability documentation at the module level and for the manufacturing process steps successfully completed.

3.3.6. The use of brushing to remove solder balls shall not be allowed in an area with SMD components.

3.3.7. To avoid migration (dendritic growth) problems for rework of SMD or THT only solder core with low halide (e.g. chlorine) content flux per J-Std-004B shall be used. Flux Pens shall not be used.

3.3.8. Audits on solder quality of reworked boards shall be done on a regular basis based on the PFMEA. The flux residues on the solder joints shall be checked with flux indicator liquids (e.g. Zestron or other brands) to avoid localized high concentration residue that leads to dendritic growth. PCBs with all allowed reworks shall be included in the evaluations.

3.3.9. All rework or repair processes in the supplier manufacturing facility shall be incorporated in the production Control Plan and the PFMEA.

3.3.9.1. Any material (fabric pieces, cloths, sponges, rags) used to clean the PCBA after rework is completed shall be disposed of after they are used (preferred practice is single use) to prevent potential contamination of the PCBA

3.3.10. There shall be no rework permitted for failure modes with a 7 to 10 severity ranking unless approved in writing by the GM Supplier Quality SMT Manager or above and the deviation shall be documented in SQMS as part of the PPAP.

3.3.10.1. Failure modes with a severity of 9 or 10 should have zero tolerance for any type of rework. Rework may be allowed, with deviation previously approved in (3.3.10). SQE shall approve the allowed reworks and it will be documented in SQMS as part of the PPAP ~~;~~ only if all conditions defined in section 3.5.23 are subsequently met (post-rework).

3.3.11. Ball Grid Array (BGA), Quad Flat – No leads (QFN), and Dual Flat – No Lead (DFN) devices shall not be reworked unless the rework process is specifically validated.

3.4. The supplier manufacturing location shall incorporate the following <u>Lessons Learned and Best Practices</u> into the *control of areas of potential electro-static discharge* (ESD) [I]

3.4.1. The supplier manufacturing location shall adhere to the requirements defined in ANSI/ESD S20.20 (latest released version) or IEC 61340-5-X lasted revision

3.4.1.1. All findings classified as “Fail”, or “Needs Immediate Action”, or “Not Satisfactory” shall be included in a continuous improvement plan and could, at the discretion of the SQE, impact Full PPAP status.

3.4.2. An annual assessment shall be completed using the GM 1927 16b Electrostatic Discharge and Electrical Over Stress Management Process Audit”

3.4.3. An internal audit shall be developed to control areas of potential risk for each non-compliance identified in the above mentioned AIAG & GM assessments.

3.4.3.1. The audit shall address both Human Body Models (HBM) and Charged Device Models (CDM)

3.4.3.2. ESD measurements shall be performed once every quarter per each step of the process.

3.4.3.3. ESD event meter measurement is required once per year on each station for 30 min.

3.4.3.4. ESD walking test measurement is required once per year on each station.

3.4.4. Procedures to support ANSI SP27.1-2018 “Recommended Information flow for potential EOS issues between automotive OEM, Tier 1, and semiconductor manufacturers” shall be in place. Assessment of compliance can be performed using the “GM1927 16b Electrostatic Discharge and Electrical Over Stress Management Process Audit” document.

3.5. The supplier manufacturing location shall incorporate the following <u>Lessons Learned and Best Practices</u> into the *manufacturing process control plan.*

3.5.1. Food and drinks in the production area shall not be allowed. Water is allowed in designated areas in a closed container.

3.5.2. No jewelry allowed by those handling appearance parts to avoid scratches. Boundary samples of faceplates, or any class A surface shall be easily available to the operator.

3.5.3. Visual aids and boundary samples for assembly (PCB level, module level, and assembly level) shall be available to the operator and approved to the latest revision with proper labeling.

3.5.3.1. All gauges and test equipment shall be provided with ‘Red’ masters (No Go Masters) which represent the most common process defects as denoted in the PFMEA.

3.5.4. Any cleaning chemicals used to clean tools, molds or equipment that may come directly or indirectly in contact with the PCB shall not contain any ionic compounds.

3.5.5. Any module sub assembly or a component which falls on the ground shall be scrapped. The drop policy shall be documented in the process control plan.

3.5.6. If Laser/Heat and Friction Welding of plastic housings are used, then a SPC leakage burst [N] test shall be done as part of the process control plan. And a 100% normal leakage test shall be done.

3.5.6.1. For components defined as sealed, positive pressure test with a minimum pressure differential of 48 kPa (7 psi) shall be implemented at End Of Line to detect leakage, the leakage rate shall be zero or as defined in the CTS/CTRS/SSTS.

3.5.7. Gloves or finger cots shall be used to avoid dirt, grease or body fluids contaminating the parts. Dirty gloves or finger cots shall be scrapped. Used gloves or finger cots shall be scrapped following established cleanliness standards. They shall be produced free of chlorides or sulfates.

3.5.8. Any ports or openings in the module case used for validation or special test purposes shall be covered with tape or label material prior to shipping to GM.
3.5.9. To ensure the underside components on the PCBA are not damaged when the opposite side is being processed (i.e. solder screen paste operation thru ICT) quality sampling checks shall be performed using an appropriate method and at a frequency determined by the process capability to prevent potential defective material from being shipped.
3.5.10. PCBs with Conformal Coating shall be cured according to coating material supplier requirements (e.g. air circulation, humidity) and any manufacturing process controls (e.g. tac free, runs).
3.5.11. Before PCB solder paste, the PCB shall be free of dust/fiber contaminants (e.g. ionized air blowers in conjunction with a vacuum system). Electrostatic cleaners or sticky tape devices can be used.
3.5.12. If screws are used in the assembly, the following shall be monitored and controlled 100%
3.5.12.1. Screw torque
3.5.12.2. Revolution angle (time or screwing depth or number of revolutions)
3.5.12.3. The number of screws.
3.5.12.4. The screwdriver shall be guided, kept vertical & perpendicular head by a mechanical arm.
3.5.12.5. The screwing fastening sequence shall be defined and verified.
3.5.12.6. Workstation shall be stable to prevent part movement during assembly.
3.5.12.7. Electric torque screwdrivers shall be electrically grounded to prevent EOS from occurring.
3.5.13. Error proof verification of high severity (7, 8, 9 or 10) failure modes shall use both OK and NOK parts. Frequency shall be at a minimum once per shift or when a different product is produced or after equipment maintenance is performed.
3.5.14. When areas for potential mechanical interference with PCB vias are identified, the supplier shall have preventative controls to ensure solder domes cannot be created.
3.5.15. If potting material is used in the assembly,
3.5.15.1. The potting pressure (sealant) shall be monitored to avoid air bubbles.
3.5.15.2. When material is replenished, the system shall be verified no air bubbles were created.
3.5.15.3. Volumetric pumps shall be used for application
3.5.15.4. Bar code labels shall be used to confirm correct potting material is used.
3.5.15.5. Regular weight shots shall be done for SPC monitoring.
3.5.15.6. For complex cases a camera AOI inspection shall be used instead of visual inspection.
3.5.16. If Conformal coating is used it shall be applied with an automated process.
3.5.17. The manufacturing process shall be set up so that there will be no excess flux or moisture residue on the PCB. PCB holding fixtures shall be configured and orientated to prevent any flux from potentially creating a short-circuit path.
3.5.18. The stacking of PCB assemblies shall not be allowed.
3.5.19. Warning labels shall identify any areas used for processing or storage of components that are sensitive to mechanical stress.
3.5.20. If internal connectors are used, then error-proofing shall be in place to ensure connectors are securely locked into mating connector [G].
3.5.21. The PCB assembly shall cool down to the specified temperature after any process involving heat (e.g. Reflow oven, solder bath, curing ovens, test chambers) before any additional processing or testing is performed.
3.5.22. Controls shall be in place to meet or exceed IPC 5703 Cleanliness Guidelines for Printed board fabricators and IPC 6012 section 3.9 & 3.10 standards for cleanliness on populated PCB assemblies to prevent dendritic growth opportunities. The controls shall be based on severity rankings of the PFMEA and include temperature and relative humidity controls.
3.5.23. For PFMEA failure modes with a severity ranking of 9 or 10, the corresponding detection ranking shall be 3 or lower (per 4th edition AIAG PFMEA Manual - table Cr3 criteria or AIAG VDA FMEA Handbook and IATF 16949 - GM Customer Specific Requirements, whichever is applicable).
3.5.23.1. PFMEA detection rankings of 4 are acceptable if either of these conditions are met:
3.5.23.1.1. There is traceability back to the original operation where the failure mode occurs
3.5.23.1.2. There are no manual operations between the detection and the original operation (e.g. automated line)
3.5.23.2. In cases where detection requirements are not met the GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition form will need to be completed.

3.5.23.3. Approved deviations to the detection level requirements are listed in Appendix E. The GM 1927 21 DFMEA PFMEA Gap Analysis Process and Transition form will not need to be completed if the controls listed are in place.

3.5.24. For PFMEA failure modes with a severity ranking of 7 or 8, the corresponding detection ranking shall be 4 or lower.
3.5.25. Only silicone which doesn't vapor out shall be used through the supply chain productions because silicon oxide can build up an isolation layer on contacts.
3.5.25.1. All silicone shall be removed from exterior surfaces as it can cause paint adhesion issues at the vehicle assembly plants.
3.5.26. All surface mount components shall be properly handled according to their moisture sensitivity classification throughout the manufacturing supply chain per IPC standard JEDEC J-STD-020 and JEDEC J-STD-033).
3.5.27. All printed circuit boards or substrates shall be properly handled throughout the manufacturing supply chain per the IPC standard 1601.
3.5.28. Bare PCB from the PCB supplier manufacturing location shall meet IPC 5704 cleanliness standards. In addition, the following requirements shall be met.
3.5.28.1. The chlorides shall be less than 1*u*g / in^2
3.5.28.2. Sulfates shall be less than 3 *u*g / in^2
3.5.29. The power (voltage and current) used in the sonic welding process should be monitored by SPC.

3.6. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the *manufacturing equipment preventive maintenance program.*
3.6.1. If a power interruption occurs during production, any material in process shall be re-certified before continuing in the process.
3.6.2. A power supply backup system should also be in place to prevent unplanned downtime to the manufacturing / test equipment programming.
3.6.2.1. Programming parameters not stored in a non –volatile memory location shall be buffered or reloaded after a power interruption.
3.6.3. The mounting of critical fixtures to production equipment shall be secured to a defined torque. The fixtures shall be inspected to detect loosening.
3.6.4. The installation of critical production equipment shall be secured by a lock-tight type mechanism and have an inspection method in place to detect loosening by vibration.
3.6.5. After In-Circuit-Tester pin / probe replacements, the equipment shall be checked to verify no damage to the PCB or components can occur. Force values shall also be confirmed.
3.6.6. All devices (fixture, cover, clamps, etc.) used to hold the PCB assembly in place during the selective soldering (or other soldering) process shall be included in the cleaning schedule.
3.6.7. Verification that underside components on the PCBA will not be damaged when the opposite side is being processed shall be included as part of any maintenance work being performed.

3.7. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the *manufacturing inspection / test equipment:*
3.7.1. For any changes to the part, process, or any test equipment including software, sample units (e.g. first article) shall be reviewed with the GM Product Development Team (DRE, SQE, CVE) to confirm compliance to the validated design record requirements. Approval shall be formally documented (e.g. GM3660, SPCR, CN)
3.7.2. Vehicle wiring harness header/connectors should not be used as part of the functional test station. Holders shall be in good condition and do not damage the components. Connectors with spring force pins (e.g. pogo pins) or radial spring force barrel type connectors (also called spring compression) shall be used.
3.7.3. Connector Terminal Pin alignment shall be checked 100% prevent shipping product with bent pins. The preferred methods are:1) AOI camera method, 2) X-Ray inspection method, 3) mechanical gage.
3.7.3.1. There shall be no additional work performed involving the connectors after this inspection.
3.7.3.2. Manual, visual only, inspection is not allowed
3.7.4. Automatic Optical Inspection (AOI), In-Circuit-Tests (ICT) and Automatic Functional tests (AFT) shall be used on 100% of production.

3.7.4.1. A Test Coverage Matrix (component existence, open-circuit, short-circuit, correct polarity, correct value/tolerance/temperature rating) shall be used to error proof that every solder joint and component is tested.
3.7.4.2. Internal connections and external connectors in the assembly (e.g. mechanical short circuits between male and female connectors) shall also be included in the matrix [G].
3.7.4.3. If components or solder connections are not visible to the AOI equipment, then additional controls shall be in place to verify compliance to requirements

3.7.5. A 2.5 Dimensional (camera at 45 degree angle) or a 3D Automatic Optical Inspection () shall be used for 100% of the solder joints/paste application for the following conditions:
3.7.5.1. After SMD Paste-Print if BGA (Ball Grid Arrays) are on the PCB, otherwise 2D AOI is OK.
3.7.5.2. If BGA devices are used, a combination of 2D AOI at Paste Print and X-ray after SMD oven is allowed.
3.7.5.3. After SMD Oven if Fine Pitch ICs are on the PCB, otherwise 2D AOI shall required.
3.7.5.4. After Wave or Selective (Pot or Fountain) soldering if no press fit or SMD is available.
3.7.5.5. If complexity on PCB is high (more than 20 Pins wave/selective soldered).
3.7.5.6. If the pitch of IC pins is smaller than 1.27 mm. If not, then ICT can be used to check all pins for short-circuits and a visual check (solder ball, shorts, and meniscus) shall be done.
3.7.6. Personnel checking the Defective/Non-conforming/Suspect-AOI parts on the monitor should rotate after 2 hours to a non-visual demanding operation to avoid increased failures.
3.7.6.1. Only IPC certified / trained personnel shall be allowed to over-ride any non-conforming or suspect parts found at solder paste inspection station (prior to pick & place of components)
3.7.7. In order to evaluate the parasitic/quiescent current draw of modules in sleep mode the supplier shall implement a “fast shut down” mode on the functional tester. The “fast shut down” function shall be able to force the part into “sleep mode” and shall measure the parasitic/quiescent current after an 8 second minimum interval in order to stabilize the circuitry.
3.7.8. The installation process for compliant (press fit) pins and any in-process testing shall adhere to the latest edition of the IEC-60352-5 standard, In addition:
3.7.8.1. An optical or mechanical control method for pin presence & position shall be implemented.
3.7.8.2. During the press-fit installation process the force and distance shall be 100% monitored and results documented by production lot code. [M]
3.7.8.3. 100% verification that the correct pin is installed
3.7.8.4. Electrical continuity of any circuit utilizing a press fit connection shall be verified
3.7.8.5. Pin samples shall be measured for pin retention (from each processing position and Final assembly) at the start-up and end of every production run using a variable measurement system
3.7.8.5.1 Data shall be plotted on a SPC Run Chart for monitoring variable drift in components or equipment. Production Lots shall be contained and released after the final assembly check identifies no non-conforming results.
3.7.9. The In-Circuit Tester (ICT) shall not check conformity of components by probing a component pad. If the technology utilizes a “bed of nails” type method, then the PCB shall have separate pads dedicated to measurement verification. If it is a “flying probe” technology, then separate pads are not required.
3.7.10. The functional tester at the supplier manufacturing site shall utilize electrical load values (impedance, inductance, capacitance, voltage) consistent with the GM vehicle application.
3.7.10.1. Sample units shall be confirmed on a GM vehicle to verify the functionality and test limits used are consistent with the results of the final test station
3.7.10.2. The natural vehicle loads shall be used in the development of the final test station to find potential “hysteresis” issues.
3.7.11. Audits on solder quality shall be done checking for through-hole solder intrusion, solder radius, and solder fillet integrity. The flux residues on the solder joints shall be checked with flux indicator liquids (e.g. Zestron or other brands) to avoid dendritic growth.
3.7.12. When areas for potential mechanical damage are identified as a result of the orientation of the part in the fixture (such as a test station cover, or a fixture clamp), the *orientation error proofing* shall ensure parts are not damaged prior to proceeding with the testing of the part.
3.7.13. If conformal coating is used, it shall be inspected with an ultra violet light method.
3.7.14. Audits on the cleanliness of contacts used in relays and switches shall be conducted on a process capability driven frequency that establishes that there is no contamination present.

3.7.14.1. Visual only inspection is not acceptable. At a minimum a 10x magnification should be utilized. A chemical analysis is preferred.

3.7.15. Verification of display illumination shall be done with an AOI vision system.

3.7.16. Functionally verify all displays by using a predetermined pattern or configuration appropriate for the type of display

3.7.17. Vibration and thermal tests shall be implemented to detect intermittent internal connection failure modes for all PFMEA failure modes identified as high severity (8, 9, or 10) or safety critical.

3.7.17.1. The test plan and equipment required shall be approved by the GM Product Development Team (CVE, DRE, and SQE) prior to sourcing. It shall include the number of samples to be tested and the specific tests performed, and equipment required.

3.7.18. The test plan and any test equipment used to evaluate functionality or measure parametric values shall be approved by the Global Sub-System Leadership Team (GSSLT).

3.7.19. The solder paste inspection equipment shall incorporate the acceptance requirements of electrical assemblies as described in the IPC-7527

3.7.19.1. Solder paste volume should be monitored with SPC

3.8. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the manufacturing process for *labeling / traceability.*

3.8.1. Any labels/stickers used on PCB's (e.g. traceability) shall only be placed where there is no potential for electro-chemical migration or contamination. (e.g. PCB traces and pads)

3.8.2. Verification of the part information on the label and what is loaded into the electronic memory of the unit shall be error proofed.

3.8.2.1. The pass-fail status of the part shall be generated from reading internal data.

3.8.2.2. All internal data and external label information shall agree.

3.8.2.3. Non-serialized labels shall not be printed prior to the finished product being tested.

3.8.2.4. FCC certification label shall meet FCC requirements and shall be traceable back to the-QR label/Assembly label before to be shipped to GM.

3.8.3. The application of a Pass/OK status shall not be completed prior to the successful completion of all required testing.

3.8.4. Traceability capability shall be in place through the entire manufacturing process (e.g. WIP, finished goods, in transit) to support suspect material containment, first-in-first-out inventory control and problem-solving activities.

3.8.4.1. Suspect lot codes of material shall be electronically identified so that they cannot be introduced into the manufacturing process at any point.

3.8.5. Non-conforming material shall be electronically statused (module based, or server based) or identified (e.g. flag, serial number, bit flip) to prevent product from being shipped to the customer.

3.8.5.1. Red marked containers (e.g. totes, trays,) shall be used for defective material [A].

3.9. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the *manufacturing process / equipment development activity:*

3.9.1. The use of under PCB supports shall be minimized so that there is no potential interference with opposite side components already placed on the PCB. Parallel supports are the preferred method.

3.9.2. Spray-Flux processes should be avoided where possible. If unable, then a drop-jet flux application is preferred over an ultrasonic application system. These flux materials shall be non-conductive/non-hydroscopic.

3.9.3. The use of glue or epoxy to hold larger components (e.g. large capacitors) in place for mechanical stress relief shall have temperature and volume dispense controls.

3.9.4. Purchased components, materials and equipment used in the manufacture of products shall be used within the guidelines and specifications of the supplier. Supplier shall notify GM Product Development Team (DRE, SQE, and CVE) prior to any changes or variations from published standards.

3.9.5. If they are used in the design of the part; Pigtails or cables exiting housings shall not be used to install or remove the part from the test fixture.

3.9.6. The use of glue to hold SMD components in place prior to soldering should be minimized. If gluing is used, they shall have temperature and volume dispense controls. Mask glue application shall not be allowed.

3.9.7. The mechanical stress (strain) affecting the PCB anywhere in the manufacturing process shall be at a level that does not damage any of the components or the integrity of the soldered connections. Any fixture that comes in contact with the PCBA shall not damage or break any of the components or soldered connections on the PCBA.
3.9.8. When BGA under-fill is required, volume dispense controls and the controls for the temperature of the PCB shall be in place. The curing oven temperature shall also be controlled.
3.9.9. All PCB arrays using High Tg (glass) laminates shall be separated using a process that minimizes mechanical stress. Routing is the preferred method to eliminate potential contamination from punching of such materials and stress induced damage to components.
3.9.10. A position locator device for the component being soldered shall be used to guarantee capable and correct position during the selective/wave soldering process.
3.9.11. Washing of Printed Circuit Boards that have been populated with components shall be avoided unless specific controls are in place and the process has been validated and approved by the GM SQE.
3.9.11.1. Cleaning with an ultrasonic bath is an acceptable practice to remove surface contamination.
3.9.12. The thermal profile of any operation utilizing heat (reflow oven, wave solder, curing, etc.) shall be optimized (time and temperature) to burn off any residual flux residue while not exposing the SMD components to any thermal degradation or solder deficiencies.
3.9.12.1. Regular calibration verification testing shall be completed for continuous running SMT profiles.
3.9.12.2. A redundant continuous monitoring program with extra sensors to record the oven temperature profile shall be in place with traceability to PCB level.
3.9.13. Reflow ovens shall have an active flux management process to remove flux residues from the furnace chamber
3.9.14. For leakage tests the vacuum chamber seal shall not be in the metal or die cast housing sealing groove.

3.10. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the *Containerization activity:*
3.10.1. The use of returnable containers should be utilized where possible.
3.10.2. If returnable containers are utilized internally, the supplier shall be responsible for cleaning the inside of the containers.
3.10.3. If returnable containers are utilized the supplier shall be responsible for removing or covering up any old labels unless GM has contracted a local third party to provide the service.

3.11. The supplier manufacturing location shall incorporate the following Lessons Learned and Best Practices into the GM 1927 28 Early Production Containment (*EPC) activity:*
3.11.1. LCD panel modules shall be 100% age checked (e.g. customized to the display 60°C 2h or 70°C 1h, not to damage). It is recommended to be continued for 100% during normal production.
3.11.2. For product on new vehicle launches, involving new hardware and / or new software, the GM 1927 28 EPC exit criteria shall include the following:
3.11.2.1. There shall be no supplier responsible warranty returns at 2 Months in Service (MIS) for 30 production days.
3.11.2.2. The first 500 parts shall undergo a “burn in” type activity unless approved by GM to use field performance data from similar product. At a minimum the burn in shall be 1 Power temperature cycle from the latest edition of GMW 3172 in order to identify infant mortality or other types of failure modes. Additional test parameters may include vibration, ignition cycling, inputs cycled, Part functionality (outputs) shall be monitored electronically.
3.11.2.3. The suppler shall formally request to the GM Product Development Team (DRE, SQE, CVE) to exit from or to modify the GM 1927 28 exit criteria (such as in the case of a corporate common part used in multiple assembly plants).
3.11.2.4. All PFMEA failure modes identified as high severity (8, 9, or 10) or safety critical, shall be clearly documented in the control plan and visible throughout the GM 1927 28 inspection process. This could include a production test station being utilized as a GM 1927 28 inspection station. (ICT, AOI, etc.)

3.12. The supplier manufacturing location shall incorporate the following Lessons Learned and Best practices into the *Compliance Documentation* activity:
3.12.1. Each shipment of PCBs to the manufacturing facility should include the following:
3.12.1.1. Cleanliness of bare PCB at the PCB supplier manufacturing location (IPC 5704)

3.12.1.2. Drilling Guidelines for PCB: copper in the via is a minimum 25um, and the hole roughness is a maximum 25um. (IPC-DR-572A)

3.12.1.3. Cross sections in via hole: first piece, last before drill. Including copper thickness and roughness. The drill bits should be changed at 80% of their lifetime usage.

3.12.2. The manufacturing facility shall maintain documentation and be available upon request for review by the SQE to show that all components used in the product meet, at a minimum, AEC Q100 (Integrated Circuits), AEC Q101 (Discrete Semiconductors), AEC Q200 (Passive Components) requirements.

3.12.3. If applicable, the manufacturing facility shall maintain documentation to show that the Bar code label requirements are being met and approval documented (CG2503).

3.12.4. If applicable, the manufacturing facility shall maintain documentation to show that the Painted plastic requirements are being met and approval documented (GMW17407 Exterior Parts, GMW17977 Interior Parts).

3.12.5. If applicable, (eSOR contains appendix F-14) the manufacturing facility shall maintain documentation to show that the requirements outlined in the “GM Third Party Information Security Requirements” have been met.

4. Quality Reliability Durability (QRD) focused Part specific requirements

The supplier manufacturing location shall incorporate any additional requirements from the following Part Specific Statements of Requirements into the manufacturing process that apply:

4.1. Electrical SMT

4.1.1. Body Electronics

4.1.1.1. Horns

4.1.1.1.1. Gap control between movable internal components shall be monitored / controlled.

4.1.1.1.2. Sealing adhesive rework shall not be allowed.

4.1.2. Chassis Electronics

4.1.2.1. Tire Pressure sensors

4.1.2.1.1. The following parameters shall be verified and recorded for each sensor during the end-of-line test:

-Correct pressure reading and proper radio frequency (RF) transmissions per the GM RF protocol.

-Output power.

4.1.2.1.2. The following inspections shall be performed on 100% of production

-Correct battery orientation and soldering of battery tags.

-Potting defects, Valve stem and valve grommet damage

-The approved grommet is correctly in place.

-The air hole has been drilled and is not obstructed or plugged.

-The valve cap is the correct type and threaded on tightly. If a plastic cap is used, it shall have no tears, cracks or other damage.

-If a lubricant or coating is put on the valve stem or grommet, correct type and amount used shall be verified and that it is applied in the proper location.

4.1.3. Telematics

4.1.3.1. Modules

4.1.3.1.1. A live GPS signal with traffic data (and not a bench generated signal) should be used to test modules with a GPS receiver

4.1.4. Infotainment

4.1.4.1. Clusters & HUDs & Displays

4.1.4.1.1. The device shall never be repaired more than 1 times for the same failure. A device shall be scrapped if it fails after 3 attempts total to repair it.

4.1.4.1.2. Appliques utilizing adhesive shall be pressed to inner housing (base board) to prevent the appliqué from lifting or buckling. DFMEA / PFMEA review for appliques assembly potentials.

4.1.4.1.3. Any fixture used to assemble a housing with snap features shall have sensors to verify all snaps are locked and engaged.
4.1.4.1.4. Lens packaging should minimize contamination/foreign debris. Any containers should have a maintenance program to ensure compliance.
4.1.4.1.5. Painted parts such as a mask/retainer must cure 72 hours at room temperature (or shorter when curing oven is used until no vapor can damage lens) before attaching to the clear plastic lens.
4.1.4.1.6. For devices with Pointers:
4.1.4.1.6.1. Pointer staking process capability studies (variable not attribute) shall be conducted and control limits (boundary samples) determined prior to PPAP approval. [J]
4.1.4.1.6.2. Pointer staking on stepper motor shall be 100% measured for proper clearance between appliqué and the pointer hub assembly
4.1.4.1.6.3. Pointers shall be calibrated after the lens is installed.
4.1.4.1.6.4. Once a pointer is installed, and prior to outer case / lens assembly, NO air hoses or forced air blow off shall be used.
4.1.4.1.7. For devices with stepper motors:
4.1.4.1.7.1. The stepper motor shall have an on-going continuous compliance testing program. The minimum testing should be once a month.
4.1.4.1.7.2. The combined stepper motor noise in an assembled device shall be monitored to confirm compliance to the design requirements.

4.1.4.2. Antennas
4.1.4.2.1. Any test involving signal strength evaluation for active antennas shall be conducted at a minimum of three points (low, mid & high) of the specification.
4.1.4.2.2. All vehicle roof mounted antennas shall be checked for all dimensions of mating surfaces by using appropriate checking device such as Go-No-Go gage.

4.1.4.3. Modules
4.1.4.3.1. A live GPS signal with traffic data (and not a bench generated signal) shall be used to test modules with a GPS receiver

4.1.4.4. Speakers
4.1.4.4.1. There shall be a controlled quantity of adhesive applied on every component joint. The quantity of adhesive shall be measured, and process capability should be demonstrated. Automated adhesive dispensing is required unless approved by the GM PDT (on lower volume projects).
4.1.4.4.2. Adhesive handling and storage shall be controlled and applied under conditions (temperature, humidity, expiration, etc.) recommended by adhesive supplier and/or previous product performance studies.
4.1.4.4.3. Open and close time of the press should be controlled. Time elapsed after the adhesive application and joint pressing shall be controlled and monitored to ensure the proper process cure, according to the specifications defined by adhesive supplier and/or previous product performance studies. Poke Yokes/Error Proofing shall be included to ensure robust process.
4.1.4.4.4. Pressing force and the pressing time should be monitored and controlled by automated device.
4.1.4.4.5. Curing time on joint shall be controlled after press operation. The joint shall not be disturbed before the curing time established, according to the specifications defined by the adhesive supplier and/or previous product performance studies. Poke Yokes/Error Proofing shall be included to ensure robust process.
4.1.4.4.6. There shall be control parameters in the curing oven to assure the correct adhesion and joints being evaluated along with process capability.
4.1.4.4.7. Contamination shall be controlled at the supplier manufacturing processes to assure the expected adhesion in the joint. Use of monitored vacuum devices shall be applied before dispensing the adhesive.
4.1.4.4.8. Tier supplier components should meet surface tension requirements to obtain the expected joints according to previous product performance studies. If the suppliers (Tier 2) of the components doesn't provide the evidence that they meet with the surface tension

requirements, then the manufacturing (Tier 1) supplier shall perform the analysis to ensure compliance.

4.1.4.4.9. In the case for use of accelerator, overlap among adhesive and accelerator shall be controlled every cycle. The overlapping of adhesive shall be evaluated, and process capability should be demonstrated.

4.1.5. Switches

4.1.5.1. Reference CG4212 GM 1927 03a Switches Controls and Column Electrical assemblies Supplier Quality Part Specific Requirements – for additional requirements

4.1.6. Electrical Centers

4.1.6.1. There shall be a vision system which verifies component markings and / or color of fuses unless error-proofing methods are in place to ensure no component mixture.

4.1.6.2. Fuses and relay insertion shall be done using an automated process capable to measure terminals insertion force to detect

4.1.6.2.1. Fuse and relay mis-assembly

4.1.6.2.2. Fork terminal / female terminal / bus bar retention force degradation due to mechanical damage

4.1.6.3. Any fixture used to assemble a housing with snap features shall

4.1.6.3.1. Have sensors to verify all snaps are locked and engaged.

4.1.6.3.2. Have housing parts alignment control features to eliminate the risk of damaging the male terminals, fork terminals and bus bars

4.1.6.4. All parts shall undergo 100% inspection to check for anomalies in crimped and welded wire connections.

4.1.6.4.1. If the inspection is done visually by the operator, the inspection shall be conducted under a magnifying lens with at least 5X magnification.

4.1.6.4.2. Sample size reduction can be requested only after three months from the start of regular production through SPCR process.

4.1.6.5. Exposed conductors from an enclosed connector assembly that impacts creepage and clearance is not acceptable.

4.1.6.6. The presence of a single loose wire strand for both crimped and welded connections is not acceptable. All parts with loose wire strands shall go to a rework station

4.1.6.7. There shall be a vision system that is capable to 100% detect the fork terminal gap variation and perpendicularity to the PCB (single fork and bus bar)

4.1.6.8. The Male Blade Stabilizer (MBS) pre-stage position must be controlled using a fixture that ensures the MBS is located 2 mm (minimum) below the tip of the terminal

4.1.6.9. Electrical Center end of line test shall be capable to comply with the relay contacts Burn in Test as specified in CG 1770 (Electrical Centers CTS)

4.2. Propulsion Systems SMT

4.2.1. Control units (Engine, Transmission)

4.2.1.1. The supplier should not reuse PCB´s which were washed after a failed paste printing

4.2.1.2. Each unit shall run through a temperature cycle test as defined by the eSOR

4.2.1.3. Each unit shall be tested under real loads with multiple ignition cycles

4.3. Electrification SMT

4.3.1. The Electrification commodity is comprised of the following components:

4.3.1.1. Battery system controls, sensing, and power modules

4.3.1.2. Inverter modules

4.3.1.3. On Board Charger modules

4.3.1.4. High Voltage Fuses

4.3.1.5. High voltage cables and bus bars

4.3.1.6. High voltage Relays / contactors / disconnects

4.3.1.7. Battery cell Interconnect boards

4.3.1.8. High Energy Hybrid coolant heaters

4.3.1.9. Electric vehicle supply equipment (EVSE) cables and connectors

4.3.2. The supplier shall retain for the life of the program documentation on the following process parameters. The list of specific parameters shall be approved by the GM SQE.

4.3.2.1. Crimping process
4.3.2.2. Ultrasonic welding process
4.3.2.3. Installation of fasteners and fittings
4.3.2.4. SMT solder process including all sub tiers

4.3.3. Statistical process control (SPC) monitoring shall be implemented on the following output parameters:
4.3.3.1. Solder paste height
4.3.3.2. Ultrasonic welding pull force
4.3.3.3. Crimp pull force
4.3.3.4. Bus-bar rivet height
4.3.3.5. Bus-bar fastener torque and angle
4.3.3.6. Coolant line fittings torque and angle

4.3.4. All ultrasonically welded circuits shall have a destructive test monitoring based on USCAR-38 test methods. For those circuits that don't have available USCAR-38 test method, the supplier shall develop their own test method and submit to GM (CVE, DRE & SQE) for approval. The new test method shall be available during Non-Saleable builds.

4.3.5. The Tier 1 supplier shall require their sub tiered suppliers of unpopulated PCB's to implement copper trace dimensional integrity.
4.3.5.1. There shall be an AQC monitoring of all copper trace dimensions in the PCB (i.e. width, thickness, etc). As a best practice, dimensions shall be measured after the last etching of chemical cleaning process.
4.3.5.2. Any chemical cleaning, such as micro-etching, that removes a thin layer of copper is strictly not allowed.
4.3.5.3. The PCB supplier shall have an inspection station in place to screen copper trace defects that are undetectable at open/short test but would cause high electrical resistance.

4.3.6. Any change in the supplier process made after Non-Saleable PPAP shall follow the Supplier Process Change Request (SPCR) guidelines and approval process. Requirement to perform PPAP resubmission in SQMS (Supplier Quality Management System) for changes due to SPCR will depend on the type of the change. The decision for PPAP resubmission will be made by GM SQE.

4.3.7. All parts shall undergo 100% inspection to check for anomalies in crimped and welded wire connections.
4.3.7.1. If the inspection is done visually by the operator, the inspection shall be conducted under a magnifying lens with at least 5X magnification.
4.3.7.1.1. Sample size reduction can be requested only after three months from the start of regular production through SPCR process.
4.3.7.2. Exposed conductors from an enclosed connector assembly that impacts creepage and clearance is not acceptable.

4.3.8. The presence of a single loose wire strand for both crimped and welded connections is not acceptable. All parts with loose wire strands shall go to a rework station

4.4. Chassis SMT

4.4.1. Placeholder for future content.

4.5. Interiors SMT

4.5.1. Placeholder for future content.

5. Assembly Plant Support

5.1. The supplier shall acquire confirmed failures from the assembly plant within 48 hours and a 5-phase corrective action plan initiated regardless of if a SPPS record is issued or not.

5.2. For product manufactured outside the GM assembly plant region, the supplier shall have the capability to perform a functional containment screen per GM 1927 17 Supplier Quality Processes and Measurements Procedure in the region of the impacted assembly plants.
5.2.1. If the hardware requires any type of rework outside the manufacturing location, then a T.W.O. is required. In addition, the rework process and facility shall be validated and approved by the GM Product Development Team (DRE, SQE, CVE)

6. GM 1927 16b Process audits

6.1. Supplier shall conduct at least annually the following GM 1927 16b Part Specific Process Audits
- 6.1.1. Electro-Static Discharge (ESD) and Electrical Over Stress (EOS) Management
- 6.1.2. Contamination Control
- 6.1.3. Control of Non-Conforming material
- 6.1.4. Electronic Assembly Manufacturing-Soldering
- 6.1.5. Electronic Components Mechanical Stress Management
- 6.1.6. Surface Mount Technology
- 6.1.7. Cyber Security Requirements for Manufacturing

7. Software

7.1. Software Approval

7.1.1. The manufacturing facility shall maintain documentation to show that the version being loaded into production has been validated & approved by the GM Product Development Team (DRE, CVE, & SQE).

7.1.1.1. Any changes to the software content loaded into a device or module shall be reviewed with the DRE and CVE per the CG3000 software delta validation plan prior to shipping material to the GM Assembly facilities.

7.1.2. A “First Article” sample containing each new software version shall be confirmed by the GM Product Development Team (DRE, CVE) [K].

7.1.2.1. An A to B comparison between the validated Software files and a first production sample shall be completed for Production Part Approval and approval documentation uploaded into SQMS.

7.1.3. Any software changes to the test or manufacturing equipment that have the potential to affect the function or performance of the vehicle shall be verified on all affected GM vehicle(s) prior to shipping material to the GM Assembly facilities.

7.1.3.1. The supplier shall submit an updated GM3660 and PSW for Production Part Approval.

7.1.3.2. The supplier shall notify the GM assembly plant(s) for the Production Trial Run. If the PTR is waived by the assembly plant, it shall be documented via e-mail.

7.2. Software APQP deliverables

7.2.1. Compliance to the Software ADV Plan shall be reviewed at each SRV / CRV gate review.

7.2.2. Any issue(s) that can impact Full PPAP status by the MV salable PPAP gate shall be reviewed by the GM Product Development Team (DRE, CVE, & SQE).

7.3. Software Version Control

7.3.1. The transfer of the correct SW, boot-loader and/or Calibration File versions from Supplier Engineering to the facility performing the programming (production, re-flash, third party, etc.) shall use the image file and check sum to verify the correct data was transferred.

7.3.2. Any process that programs software content shall utilize a fixture to mechanically lock down the product until the programming is completed.

7.3.3. Any process that programs software content shall verify that the software contained no errors at a minimum, the check-sum method and an image file verification shall be used.

7.3.4. When software is stored on more than one server, updates shall be verified for ALL servers

7.3.5. If multiple locations are used to program software, a process shall be in place to assure that all usage locations have the latest version of software required

7.3.6. The internal electronic information (part number, software version) shall be used to generate the outside printed label information.

7.3.7. All error proofing and version control activity shall use the GM part number.

7.3.8. All internal data shall be verified to match data displayed on external label.

7.4. Software Cyber Security Programming

7.4.1. The supplier manufacturing location shall have controls in place to confirm that the following cybersecurity requirements identified in Appendix F-14 of the eSOR are met. [Q]

7.4.1.1. ***Secure Unlock*** for Global A Programs: Each individual ECU shall be confirmed to have been programmed with a unique (non-duplicate) seed / key pair generated by the supplier using a deterministic random number generator.

7.4.1.1.1 All ECU's shall have SBATs (Signature Bypass Authorization Tickets) removed while ECU is in transit between any facilities.

7.4.1.2. ***Key Provisioning*** for Global B Programs (with *Integrated Security Peripheral*):

7.4.1.2.1. Access to the GM *ECU Provisioning Portal* Website shall have specific individuals and their back-ups identified by name.

7.4.1.2.2. Data sets comprised of the *ECU_ID,* the *UNLOCK_ECU_KEY* with *its associated M1, M2, & M3* messages, and the *MASTER_ECU_KEY* with *its associated M1, M2, & M3* messages shall be downloaded from the *GM Provisioning Portal* at a frequency and quantity to support contracted capacity (SDC) requirements. [R]

7.4.1.2.3. Data sets shall not be separated and regrouped into new data sets.

7.4.1.2.4. Each individual ECU shall be confirmed to have been programmed with a unique data set from GM to prevent data sets from being used in additional ECU's.

7.4.1.2.5. The ECU_ID shall be associated to a supplier generated serial number and traceability records maintained such that 2-way cross-referencing is available.

7.4.1.2.6. ECU's with duplicate *ECU_ID's* shall be scrapped, and the list of ECU_ID's sent to GM (cys@gm.com) monthly.

7.4.1.2.7. The *ECU_ID &* the *M4 & M5* response messages shall be automatically communicated to GM via the ECU *Provisioning Portal*.

7.4.1.2.8. M4 & M5 response messages shall be received from the *GM Provisioning Portal* at a frequency to support contracted capacity (SDC) requirements.

7.4.1.2.9. The ECU label should not be printed and applied to the module until the *M4 & M5* response messages have been verified to be correct.

7.4.1.2.10. A *Local Verification Utility Tool* should be temporarily available as a backup for the production of modules in case Internet connectivity is interrupted. GM verification thru the *Provisioning Portal* is still required.

7.4.1.2.11. ECU's shall not be shipped to GM until *M4 & M5* response messages are verified by GM thru the *Provisioning Portal*.

7.4.1.2.12. ECU's that fail *M4 & M5* verification from GM shall be scrapped, and the list of ECU_ID's sent to GM (cys@gm.com) monthly.

7.4.1.2.13. The *M1, M2, M3* messages shall be deleted (destroyed)[S] from all supplier IT infrastructure after the *M4 & M5* response messages have been verified by GM daily.

7.4.1.2.14. *Unused Security Peripheral memory slots (KEY_2 thru KEY_n, where n=specific quantity)* shall be confirmed to be the correct value per GB4007 for all controllers delivered to GM Assembly facilities and to GM Customer Care & Aftersales.

7.4.1.2.15. The *Authoritative Counter* shall be confirmed to be the correct value per GB4007 for all controllers delivered to GM Assembly facilities and to GM Customer Care & Aftersales

7.4.1.2.16. The Key Slot Provision State Flag shall be confirmed to be the correct value per GB4007 for all controllers delivered to GM Assembly facilities and to GM Customer Care & Aftersales

7.4.1.2.17. The counter value (i.e. CID) shall be confirmed to be the correct value per GB4007 when generating M1, M2, and M3 messages for purposes of loading security peripheral key slots.

7.4.1.2.18. The ECU_IDs & M1-M5 messages shall be obscured and non-transferable from the Key Provisioning station at the supplier manufacturing line (i.e. operator HMI operation screen, etc.).

7.4.1.2.19. The primary Key Provisioning station at the supplier manufacturing line shall be locked and password protected.

7.4.1.2.20. The Key Provisioning executable file shall be read-only at operator level.

7.4.1.2.21. The ECU_ID and Serial Number shall not be overwritten during any ECU manufacturing and/or post manufacturing processes.

7.4.1.2.22. Individual ECU's tested for the vectors of KEY 2 thru Key_n, where n=total number of key slots per GB4007 shall be supported by the micro (10 or 20 or 50).

7.4.1.2.23. The response to DID F081 shall be checked per GB4007 for each individual ECU after provisioning.

7.4.1.2.24. The supplier's EOL/final tester shall verify the ECU_ID & Serial Number are written in the module and confirmed before proceeding with EOL/final testing process.

7.4.1.3. ***Key Provisioning*** for Global B Programs (<u>without</u> *Integrated Security Peripheral*):

7.4.1.3.1. Access to the GM ECU *Provisioning Portal* Website shall have specific individuals and their back-ups identified by name.

7.4.1.3.2. Data sets comprised of the *ECU_ID and UNLOCK_ECU_KEY* shall be downloaded from the GM *Portal* at a frequency and quantity to support contracted capacity (SDC) requirements. [R]

7.4.1.3.3. Data sets shall not be separated and regrouped into new data sets.

7.4.1.3.4. Each individual ECU shall be confirmed to have been programmed with a unique data set provided by GM to prevent a data set from being used in additional ECU's.

7.4.1.3.5. The ECU_ID shall be associated to a supplier generated serial number and Traceability records maintained.

7.4.1.3.6. ECU's with duplicate *ECU_ID's* shall be scrapped and the list of ECU_ID's sent to GM (<u>cys@gm.com</u>) monthly

7.4.1.3.7. The *UNLOCK_ECU_KEY* shall be deleted (destroyed) [S] from all supplier IT Infrastructure daily.

7.4.1.3.8 The ECU_IDs & & UNLOCK_ECU_KEY data sets shall be obscured and non-transferable on the provisioning station at the supplier manufacturing line (i.e. operator HMI operation screen, etc.).

7.4.1.3.9 The primary provisioning station at the supplier manufacturing line shall be locked and password protected.

7.4.1.3.10 The provisioning executable file shall be read-only at operator level.

7.4.1.4. ***Manufacturing Enable Counter*** *(MEC)*:

7.4.1.4.1. The *MEC* shall be confirmed to be the correct value per GB4007 for all controllers delivered to GM Assembly facilities and to GM Customer Care & Aftersales.

7.4.1.5. ***Controller (ECU) Hardening***:

7.4.1.5.1. The *JTAG* port of each ECU shall be locked with a unique *PEPU* password.

7.4.1.5.2. The ECU specific *JTAG* password shall be destroyed from all supplier IT infrastructure after provisioning into the ECU. [S]

7.4.1.5.3. The JTAG Master password shall be retained for use with the *Password-Based Key Derivation Function 2* password utility program to recreate an ECU specific password.

7.4.1.5.4. All supplier test Software shall be erased from the ECU before shipment to GM.

7.4.1.6. Key Provisioning for SDV - Flexible Security Key Architecture (FSKA)

7.4.1.6.1. Access to the GM ECU Provisioning Portal Website shall have specific individuals and their back-ups identified by name. At least one individual with access to the ECU provisioning portal should be available during production of the product. The access needed is for the M2M interface, which requires OAuth2 Client Credentials. GM IECS Web Portal is accessible via https://gmsupplypower.covisint.com/web/portal
Note: Suppliers need to click on "IECS Portal" link from GM Supply Power =>COLLABORATE =>Applications

7.4.1.6.2. Data sets comprised of the default ECU_ID, the default UNLOCK_ECU_KEY with its associated M1, M2, & M3 messages, and the default MASTER_ECU_KEY with its associated M1, M2, & M3 messages shall be downloaded from the GM Provisioning Portal. All ECUs of a particular product shall be delivered to GM with the same default ECU_ID.

7.4.1.6.3. The supplier shall treat the default MASTER_ECU_KEY M1, M2, M3 messages and default UNLOCK_ECU_KEY M1, M2, M3 as confidential information that is only accessible to those who need the data

7.4.1.6.4. The supplier shall implement a detection level 3 process control as defined in CG4338 GM 1927 03 for the default MASTER_ECU_KEY provisioned into the ECU during the ECU manufacturing process is part of the matched set of default ECU_ID and default UNLOCK_ECU_KEY as provided by GM.

7.4.1.6.5. All ECUs supporting CYS2407 – APS Regional Key Provisioning Specification shall be supplied to GM with the ECU ID set to the product-specific default ECU ID provided by GM.

7.4.1.6.6. Production/fielded microcontrollers shall not contain any method to be able to provision, reprovision, change or erase the ECU_ID and ECU ID Extension in the microcontrollers other than what is specified in CYS2407 and GB6002.

7.4.1.6.7. The supplier shall ensure that the setting of the initial CyS Region Info has successfully occurred for all ECUs delivered to GM by performing the following checks:

ECU ID read via DID $F0F3 = Default ECU ID provided by GM

ECU ID Extension read via DID $E0B2 > 0 and < 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

Root Public Key hash read via DID $E0B3 = Default Root Public Key Hash value provided by GM

APSRKP Region read via DID $E0B4 = 0x0F

APSRKPC read via DID $E0B4 = 0

7.4.1.7. FSKA Micro controllers without security peripheral

7.4.1.7.1. If a design does not require support of a security peripheral (ISPKSP = 0x00 and NOKSS = 0x00)but will support the GM UDS Security Access service, the supplier shall provision the GM provided default Master_ECU_Key and UNLOCK_ECU_KEY into the secure static non-volatile memory (NVM) of the microcontroller by providing the default UNLOCK_ECU_KEY M1-M3 messages to the ECU.

7.4.1.7.2. ECUs which do not have a Security Peripheral are required to verify the M4 and M5 messages resulting from loading the default MASTER_ECU_KEY and default UNLOCK_ECU_KEY by using the local verification utility provided by GM

7.4.1.8. FSKA Microcontrollers with security peripheral

7.4.1.8.1 For ECUs containing a microcontroller which does support a security peripheral (ISPKSP = 0x01 or NOKSS > 0x00), MASTER_ECU_KEY and UNLOCK_ECU_KEY shall be initialized/provisioned to zero key (16 bytes of zeros) before provisioning GM provided default MASTER_ECU_KEY and default UNLOCK_ECU_KEY.

7.4.1.8.2 Supplier shall generate M1, M2 and M3 for provisioning zero keys (16 bytes of zeros) in any key slot. To ensure that zero key provisioning was successful suppliers are required to check M4 and M5 returned by the security peripheral.

7.4.1.8.3 The ECU supplier shall load the GM provided default Master _ECU_KEY AND UNLOCK_ECU_KEY into the security peripheral (ISPKSP = 0x01 or NOKSS > 0x00) by passing the M1, M2, M3 messages to the security peripheral using the specific security peripheral interface for loading keys.

7.4.1.8.4 Suppliers responsible for ECUs which have a Security Peripheral are required to verify the M4 and M5 messages resulting from loading the default MASTER_ECU_KEY and default UNLOCK_ECU_KEY by using the local verification utility provided by GM.

7.4.1.9. FSKA KP KEYS

7.4.1.9.1 The supplier shall treat the KP UID and KP Key sets generated by the KP Key Provisioning API as confidential information that is only accessible to those who need the data (e.g., for provisioning parts at the supplier manufacturing facility).

7.4.1.9.2 A KP Key Provisioning Data File provided by the KP Key Provisioning API shall only be used on one instance of an ECU (i.e. no two ECUs may be provisioned with the same set of KP Key Provisioning Data).

7.4.1.9.3 KP UID and KP Keys provisioned to the ECU shall not be separated or regrouped into new Data sets. A mismatch of the KP UID and KP Keys results in an ECU which is not usable by GM.

7.4.1.9.4 ECUs shall only be provisioned with KP UIDs and KP Keys that are provided by the KP Key Provisioning API.

7.4.1.9.5 At supplier manufacturing facilities all ECUs that support APSRKP shall be provisioned a KP Key for each region that the ECU is required to support along with a single KP UID.

7.4.1.9.6 The supplier shall ensure that the provisioning of the KP UID and KP Keys has successfully occurred for all ECUs delivered to GM by performing the following checks:
KP UID read via DID $E0B1 = KP UID provided by KP Key Provisioning API
U198C DTC (Primary Security Information Missing) is not set

7.4.1.9.7 No two ECUs ever delivered to GM shall be provisioned with the same
KP Record provided by the KP Key Provisioning API.

7.4.1.9.8 ECUs that support APSRKP are required to be provisioned with a default ECU_ID, one or more default Secret Keys, ECU-specific KP UID, and KP Keys during the ECU manufacturing process.

7.4.1.9.9 After the KP UID and KP Keys have been stored to the CyS Info Block they shall be verified as part of the “Provisioning of KP UID and KP Keys” RID 0x0559

7.4.1.9.10 Regardless of the ECU use case or the process via which the supplier received the Encrypted KP Key Provisioning Information associated with the KP UID the supplier shall destroy all copies of the Encrypted KP Key Provisioning Information (e.g., from all supplier IT systems) after production of the ECUs is complete.

7.4.1.9.11 Only ECUs that return a value for the Verification Data that matches the Verification Data in the KP Key Provisioning Data File provided by the KP Key Provisioning API shall be provided to GM

7.4.1.9.12 The supplier shall store the following information for a minimum of 20 years for each ECU delivered to GM and the information shall be made available to GM if requested.

- ECU Serial Number
- KP UID
- KP Verification Data provided by the KP Key Provisioning Tool
- KP Verification Data returned by ECU
- Root Public Key hash

- ECU ID
- Master Key M4
- Master Key M5
- Unlock Key M4
- Unlock Key M5
- ECU Trace Data Values (defined in GMW15862)

8. Semiconductors

8.1. Sub Tier Sourcing

8.1.1. A supplier self-assessment / audit (IATF 16949 Standards self-assessment) shall be conducted for all suppliers of electronic components listed in the "Bill of Material" prior to award of business.

8.1.2. An annual audit / review shall be conducted on any sub tier supplier that has had latent defect issues detected by GM assembly plants or in the vehicle.

8.1.3. The supplier shall incorporate performance data and quality history into the sourcing decision process of the semiconductor supplier(s). Supplier locations with known ongoing quality issues shall not be sourced.

8.1.4. The supplier should, where possible, purchase the semiconductor directly from the supplier of the device and not a distributor.

8.1.5. If a component is used in a safety critical circuit or has DFMEA failure mode(s) with a severity ranking of 8, 9, or 10, then the tier 1 supplier shall have a supply chain management process that is approved by the GM SQE, the DRE, and the buyer and the plan is in place.

8.1.6. The Tier 1 supplier shall meet the requirements outlined in the JEDEC 243 **that** is intended to prevent counterfeit electronic parts from entering the supply chain.

8.1.6.1. The manufacturing facility shall maintain documentation to show that all components used in the product meet this requirement

8.1.6.2. The Tier 1 supplier shall confirm that all sub tier suppliers including any third-party distributors in the supply chain comply with this requirement.

8.1.6.3. Any components purchased thru a distributor shall be identified on the "*Bill of Material for Electronic Components"* along with the actual Original Component Manufacturer.

8.2. Bill of Material database

8.2.1. Evidence of the GM PDT (DRE, SQE, & CVE) approval should be included in the PPAP documentation (Customer Specific Requirements) to show that the "***Bill of Material for Electronic Components***" has been reviewed. Failure to provide this documentation could impact Full PPAP status.

8.2.1.1. The "Bill of Material for Electronic Components" shall be available upon request by the SQE.

8.2.2. The supplier shall notify GM Supplier Quality in writing within two calendar days (48 hours) after receiving notification from semiconductor supplier of any potential quality or delivery issue.

8.2.2.1. Each manufacturing location shall include this requirement to notify GM Supplier Quality within 48 hours in their **GM** Fast Response procedures.

8.2.2.2. Failure to comply may result in SPPS(s) records being issued which could impact Sourceable IATF **16949 Standards** level.

8.2.3. The supplier shall include in the Non-disclosure Agreement (NDA) wording to authorize GM Supplier Quality to be able to contact the semiconductor supplier(s) directly and the semiconductor supplier to share information directly with GM in order to assess and manage risk to GM regarding a quality or delivery issue.

8.2.3.1. GM shall be updated directly on any information and action taken in regard to the quality or delivery issue being tracked. The impacted tier 1 supplier(s) shall continue to be responsible for managing containment and corrective action.

8.2.4. The GM GPSC semiconductor database, using CG4646 as the template, has been updated.

8.3. Performance & Validation

8.3.1. The manufacturing facility shall maintain documentation and be available upon request for review by the SQE to show that all components used in the product meet the performance requirements for:

8.3.1.1. AEC Q100 (Integrated Circuits)
8.3.1.1.1. For Copper (Cu) wire bonded components, AEC Q006 must be completed.
8.3.1.2. AEC Q101 (Discrete Semiconductors)
8.3.1.2.1. For Copper (Cu) wire bonded components, AEC Q006 must be completed.
8.3.1.3. AEC Q200 (Passive Components)
8.3.1.4. AEC Q102 (Discrete Optoelectronic Semiconductors)
8.3.1.5. AEC Q103 (Sensors)
8.3.1.6. AEC Q104 (Multichip Modules (MCM)s)

8.3.2. For any change involving a semiconductor component after Lead program saleable PPAP Approval, the manufacturing facility shall maintain documentation to show approval for:

8.3.2.1. The requirements outlined in GMW 16552 (referenced in GMW 3172) have been met. (An updated newly signed GM3660 is the preferred method for evidence of approval.)
8.3.2.1.1. The use of the Part Substitution Matrix (available on Supply Power) may be utilized if approved by GM Validation as part of the validation plan.

8.3.3. A documented plan shall be in place to monitor failure data including the first 60 days of service in the Vehicle for any component that is considered AEC Q100 (Integrated Circuits) or AEC Q101 (Discrete Semiconductors).

8.3.3.1. If the component is used in a Safety Critical circuit, then the plan shall include traceability information in order to identify and potentially recall any suspect material.
8.3.3.2. If the component has a PFMEA failure mode(s) with a severity ranking of 8, 9, or 10, the plan shall include the ability to identify where all suspect material is if failed parts are detected outside the manufacturing location.

8.3.4. The semiconductor supplier(s) shall provide to the tier 1 supplier evidence that all AEC Q100 Integrated Circuits and AEC Q101 Discrete Semiconductors have DPAT (Dynamic Part Average Test), or SPAT (Static Part Average Test) limits, approved by the PDT, and are established for all their devices (both analog and digital). This requirement applies to all electrical parameters in the manufacturing process (wafer, back end/final) as defined in Appendix 2 of the AEC Q001(Guidelines for Part Average testing)

8.3.4.1. Failure to provide or maintain this documentation could impact Full PPAP status.
8.3.4.2. In developing the PAT limits the Sub tier supplier should take into account the variation from tester to tester and load-board to load-board. (AECQ001)
8.3.4.3. The DPAT limits shall be +/- 6 Sigma from the mean of the individual wafer within a wafer lot and / or +/- 6 Sigma from the mean of an individual package lot. (AECQ001)
8.3.4.4. The SPAT limits shall be +/- 6 Sigma from the mean of the long-term historical average. (AECQ001)
8.3.4.5. DPAT & SPAT limits shall be within any specification (data sheet) limits. (AECQ001)
8.3.4.6. An example showing specification (data sheet) limits, PAT limits and Outliers is shown in (AECQ001 Appendix 1).

8.3.5 AEC Q002 Guidance for statistical yield analysis. After 6 months of production yields are updated statistical yield limits (SYL) and statistical bin limits (SBL)

8.3.6 AEC Q003 Guidance for characterization of integrated circuits. A characterization plan is on place per 3.4 Criteria for Characterization

8.3.7 AEC Q004 Automotive Zero Defects. Semiconductor supplier has a strategy plan to detect and reduce defects. Safe launch plan is on place for ramp up of the product.

8.3.8 AEC Q005 PB-FREE Test Requirements. Semiconductor supplier provide evidence per AEC Q005.

8.4 PCN (Product/Process Change Notification by electronic product suppliers or Customer Notification of Product and / or Process Change (PCN) of Electronic Components specified for Automotive Applications

8.4.1 Semiconductor suppliers submit a PCN (Process/Change Notification) per J-STD-046 or ZVEI to Tier 1/GM

8.4.1.1 Tier 1/GM start the approval process
8.4.1.2 Tier 1/GM defined the validation process
8.4.1.3 Tier 1/GM approved the PCN and PPAP is approved
8.4.1.4 Semiconductor Supplier implement a PCN and notified to Tier 1/GM first shipment
8.4.1.5 Tier 1/GM implemented the PCN and implement a break point

8.5 Product termination notifications (“PTN), **Product Discontinuance (PD) Notification Standard for product discontinuance**

8.5.1 Semiconductor suppliers submit a PD per J-STD-048 or ZVEI to Tier 1/GM

8.5.1.1 Tier 1/GM start the approval process

8.5.1.2 Tier 1/GM defined the validation process

8.5.1.3 Tier 1/GM approved the PD and PPAP is approved

8.5.1.4 Semiconductor Supplier implement a PD and notified to Tier 1/GM first shipment

8.5.1.5 Tier 1/GM implemented the PD and implement a break point

9. Appearance and Lighting Requirements

9.1. Final lighting quality shall be 100% inspected by vision tester and marked (coloring or stamping).

9.1.1. All boundary samples used for lighting checks in the manufacturing process shall be approved by the GM TALC lead(s).

9.1.2. All samples shall be stored in a similar way they would be shipped out (e.g. box of cells, and foam layers to prevent damage).

9.1.3. The lighting boundary samples shall be replaced and recertified after every engineering change that affects lighting or at a minimum of once a year (unless reviewed and approved by GM SQE).

9.2. All appearance checks (color, touch/feel, graining) shall be made against a GM approved master plaque [P]

9.2.1. All boundary samples used for appearance checks (e.g. color, feel, graining) in the manufacturing location shall be approved by the GM AAR lead(s).

9.2.2. All samples and plaques shall be stored in a similar way they would be shipped out (e.g. box of cells, and layers of foam to prevent damage).

9.2.3. The boundary samples shall be replaced and recertified after every engineering change that affects appearance or at a minimum of once a year (unless reviewed and approved by GM SQE).

9.2.4. There shall be an ongoing compliance evaluation of the appearance requirements. (E.g. Utilizing a Macbeth light or similar equipment with proven capability). [O]

9.2.4.1. The minimum frequency shall be once per shift or based on volume at all critical operations that handle decorative parts.

Appendix A – Revision History

Rev	Date	Remarks	Responsible	Approver	Approving Organization
A (1)	3/3/10	Latest version of Radio & Displays, Amplifiers, XM/OnStar, Vehicle Information Displays (VID), Electronic Modules and Electronic Safety	John Mason Martin Beck	Damon Salisbury	Supplier Quality
B (2)	3/1/14	Changed Title, Updated requirements, rearranged sections, added Commodity and Validation specific sections, and updated links to additional documents.	John Mason Martin Beck	Damon Salisbury	Supplier Quality
C (3)	11/1/14	Revised Section VII: 9-Switches to include new additional SOR reference. Added Section IX: 2 reference to new Cyber Security SOR. Added section III item 21 regarding solder fluxes. Added AECQ101 to section 1 item 17. Updated rev level of IPC 610 & 7711/7721 various places. Added item 37 & 38 to Section II regarding mounting of fixtures and equipment.	John Mason Martin Beck	Damon Salisbury	Supplier Quality
D(4)	8/14/15	Reformatted and grouped similar items and modified several sections. Revised IPC & AIAG requirements and moved them to section 2. Moved signature requirements to appendix A. Added recent lessons learned from GM global read across activity. Moved footnotes to Appendix B.	John Mason Martin Beck	Damon Salisbury	Supplier Quality
E (5)	11/20/15	1. Revised wording of items: 1.3, 3.2, 3.3, 3.3.8, 3.5.9, 3.5.20, 3.7.11, 3.8.1, 3.8.2.3, 3.8.5.1, 3.9.1, 3.11.2.2, 3.11.3, 3.11.3.1, 4.1.3, 7.1, 7.2.1, 7.3.1, 7.3.3, 7.5.5, 7.5.6 2. Removed item 1.31 3. Added items 3.3.9 & 3.3.10, 3.5.21, 3.5.22, 3.5.23, 3.5.24, 3.7.14, 3.7.15,3.7.16, 3.8.4.1, 3.12.2, 3.12.3, 3.12.4, 3.12.5,4.1.4.1.2 thru 4.1.4.9, 4.1.6, 4.2, 7.1.5, 7.5.1, 7.5.4, 4. Combined 3.5.20 with 3.7.8 5. Re-numbered items in section 7.5 6. Moved section 8 to Appendix B 7. Removed supplier signature page. 8. Moved Section 9 to Appendix A. 9. Updated Appendix C added Appendix D	John Mason Martin Beck	Damon Salisbury	Supplier Quality
F(6)	3/1/16	1. Added items 3.3.11, 3.7.17, & 5.2.1. Removed item 3.3.4.4 2. Revised: 3.3.1, 3.3.2, 3.3.10, & 4.1.5.1 3. Renumbered items in 4.1.4.1	John Mason Martin Beck	Damon Salisbury	Supplier Quality
G (7)	8/5/16	1. Updated section 3.1, 4.3, 7.4, 7.5, 2. Updated Appendix B 3. Updated item 1.3, 3.2.2, 3.3.2, 3.3.4, 3.3.9, 3.3.10, 3.3.11, 3.4.2, 3.4.3, 3.5.12, 3.5.13, 3.7.17, 3.9.2, 3.9.4, 4.1.2.1, 7.12, 7.15 4. Added footnote M, and corrected format throughout document. 5. Changed Propulsion Systems (section 4.2) 6. Added section 8 Semiconductors	J. Mason M. Beck	Damon Salisbury	Supplier Quality
H (8)	5/1/17	1. Added 3.3.9.1 2. Updated section 3.4 ESD requirements 3. Added Items 3.5.25, 3.5.26, & 3.5.27 4. Updated item 3.5.9 & 3.6.7 5. Added section 3.7.18 test plan approval 6. Updated section 4.3 Electrification SMT 7. Updated and added content to section 7.4 8. Updated 8.1.4.1 to be 8.1.5 & 8.2 & 8.3.2.2 9. Added Section 9 Lighting and Appearance 10. Added footnotes O, P, Q & R to appendix C	J. Mason M. Beck	Damon Salisbury	Supplier Quality

I (9)	August 1, 2017	1. Added additional IPC standards to section 2.1 2. Added item 3.5.28 regarding PCB cleanliness standards 3. Added item 3.7.6.1 regarding solder paste inspection operators 4. Added item 3.7.19 regarding solder paste inspection standards 5. Revised item 3.8.1 regarding label contamination 6. Removed Martin Beck as author	John Mason	Damon Salisbury	Supplier Quality
J (10)	January 18, 2018	1. Added additional references to section 2.1 2. Added item 3.5.12.7 regarding grounding screwdrivers 3. Updated sections 3.7.3 & 3.7.8 regarding compliant (press fit) pins 4. Added wording to item 3.17.19 regarding Solder volume SPC 5. Revised wording item 3.9.7 regarding soldered connections 6. Revised wording item 3.9.11 regarding washing of PCBA's 7. Revised wording item 3.9.12 regarding sources of heat 8. Added content and revised wording in section 4.3 Electrification SMT commodity specific 9. Added section 4.4 Chassis SMT commodity specific requirements 10. Revised wording item 7.4.2.1 & 7.4.2.5.1 regarding cyber security 11. Added item 8.1.6 regarding counterfeit SMD components 12. Updated item 8.2.1 regarding bill of material database 13. Added items 8.2.2 & 8.2.3 and 8.2.3.1 regarding notification of tier 2 semiconductor issues 14. Added items to section 8.3.1 regarding AEC Q006 15. Added section 8.3.4 regarding SPC limits 16. Added content to Appendix B relative to semiconductors	John Mason	Damon Salisbury	Supplier Quality
11	July 1, 2018	1. Updated section 2 & 3.2 regarding IPC610 2. Added 3.4.4 regarding EOS analysis 3. Added 3.5.29 regarding sonic welding power monitoring 4. Update section 6 regarding additional Process audits 5. Updated index page numbers in appendix D 6. Replaced GP references with GM 1927 numbers in items 3.11, 5.2 & 7.5.2.8 7. Added two items to 3.5.23 regarding PFMEA detection rankings 8. Added appF to capture 1927 17 identified exceptions 9. Added reuse & recovery to glossary (appendix B)	John Mason	Damon Salisbury	Supplier Quality
12	Oct 1, 2018	1. Updated section 3.7.4 & 3.7.4.3 regarding AOI 2. Updated section 7.1.2 & 7.1.3 regarding software 3. Updated section 7.2 regarding APQP deliverables 4. Updated section 7.4 regarding Cyber Security 5. Updated glossary for cyber security terminology	John Mason	Damon Salisbury	Supplier Quality
13	Feb 1 2019	1. Updated item 7.1.2 regarding software approval 2. Added items 7.4.1.2.14,15,16,17 regarding cyber security 3. Updated item 7.4.1.4 regarding cyber security 4. Updated section 7.4.1.5 regarding cyber security 5. Added item 8.3.2.1.1 regarding substitution matrix 6. Revised item 8.3.4 regarding PDT approval 7. Revised footnote K regarding software approval	John Mason	Damon Salisbury	Supplier Quality
14	Apr 18 2019	1. Revised references to BIQ-S and SQ Systems in items 3.3.1, 7.5.2.3, 7.5.2.9, 8.1.1, 8.2.2.2	Lidia Natanail	Damon Salisbury	Supplier Quality
15	May 21 2019	1. Removed section 7.5	Damon Salisbury	Damon Salisbury	Supplier Quality

16	Aug 1, 2019	1. Updated section 7.4 regarding Cyber Security	Arthur Fung	Damon Salisbury	Supplier Quality
17	Jan 17, 2020	1. Updated section 4.1.6 regarding Electrical Centers	Gildardo Mariano	Damon Salisbury	Supplier Quality
18	Nov 20, 2020	Updated 3.5.25 regarding silicon used through the supply chain Added 4.1.4.4 Speakers Updated 7.4.1.2.13 regarding M- messages Updated Appendix D, removed section 7.5 verbiage	Arthur Fung	Gildardo Mariano	Supplier Quality
19	Mar 31, 2021	Updated 3.12.4 regarding paint requirements Updated 8.2 regarding GPSC CG4646 requirement Updated Appendix B	Arthur Fung	Gildardo Mariano	Supplier Quality
20	Nov 15, 2021	Updated 2.2, GMW17407 Updated 3.1, further clarity to CQI 17 solder assessment Updated 3.3.10.1 further clarity to reworks requirements Updated 3.5.23, added or AIAG PFMEA Standards – GM Customer Specific Requirements GM Detection Table wording Deleted 3.11.3, continuous compliance testing Updated 7.4.1.2.5, added 2-way cross-referencing is available wording Updated GM logo	Damon Salisbury Omar Cooper Hector Orta Jun Hyun	Gildardo Mariano	Supplier Quality
21	Feb 23, 2022	Updated 2.1 External Standards/Specifications. Added 8.3.1.4 AEC Q102 (Discrete Optoelectronic Semiconductors) Added 8.3.1.5 AEC Q103 (Sensors) Added 8.3.1.6 AEC Q104 (Multichip Modules (MCM)s) Added 8.3.5 AEC Q002 Guidance for statistical yield analysis. Added 8.3.6 AEC Q003 Guidance for characterization of integrated circuits Added 8.3.7 AEC Q004 Automotive Zero Defects. Added 8.3.8 AEC Q005 PB-FREE Test Requirements. Added 8.4 PCN (Product/Process Change Notification by electronic product suppliers	Leopoldo De La Paz	Gildardo Mariano	Supplier Quality
22	Feb 28, 2023	BIQS was replaced for IATF 16949 - 3.3.1, 8.1.1, 8.2.2.1 , 8.2.2.2 Updated 3.1.2.1 and 6.1.1 16b Electronic Assembly Manufacturing-Soldering Process Audit Updated 3.3.2.2 From maximum from rework attempts to 1attempt Updated 3.3.10.1 SQE Approval to TWO approval Updated 7721/7711 3.3.3.1, 3.3.3.2, 3.3.3.3, 3.3.3.4, 3.3.4.1, 3.3.4.2, 3.3.4.3 Updated 3.4.2 and 6.1.1 from Electrostatic Discharge Management Process Audit to Electrostatic Discharge and Electrical Over Stress Management Process Audit" Added 3.4.3.2 ESD measurements needs to be performed once every quarter per each step of the process. Added 3.4.3.3 ESD event meter measurement is required once per year on each station for 30 min. Added 3.4.3.4 ESD walking test measurement is required once per year on each station. Added 3.5.22 IPC 5703 Cleanliness Guidelines for Printed board fabricators Added 3.8.2.4 Certification label shall be following the FCC requirements and marry with QR label/Assembly label before to be shipped to GM. Added 8.4 ZVEI Customer Notifications of Product and /or Process Changes (PCN) of Electronic Components specified for Automotive Applications Added 8.5 Product discontinuation Added 8.3 AECQ001	Leopoldo De La Paz	Gildardo Mariano	Supplier Quality

23	May 17, 2023	Added 7.4.1.6 Key Provisioning for SDV - Flexible Security Key Architecture (FSKA) Added 7.4.1.7 FSKA Micro controllers without security peripheral Added 7.4.1.8 FSKA Microcontrollers with security peripheral Added 7.4.1.9 FSKA KP KEYS	Karla Gamboa Leopoldo De La Paz	Gildardo Mariano	Supplier Quality

1. Appendix B – Definitions & Acronyms

- **CCA**: Customer Care & Aftersales (GM Service organization)
- **CCT:** Continued Compliance Testing: A process to validate the effect of variation in the product due to minor changes in the manufacturing process and purchased components. It is typically a sub-set of the full validation test plan.
- **Rework:** The act of correcting a defective item (using original or equivalent process) that assures full article compliance to the specifications.
- **Repair:** The act of restoring a defective item to function, and not necessarily conform to the specifications.
- **Reflash/Reprogram:** The process of overwriting/replacing of existing software (in an electronic module) with new software.
- **Reuse:** The act of recovering components from a noncomplying article thru an approved recovery process and reprocessing recovered components for a limited number of times, through the use of original processing, in a manner that assures full compliance of the article with applicable drawings or specifications.
- **Recovery:** The act of disassembling a noncomplying article to recover components so the components can be sent back to the start of the original process. A recovery is one that is done frequently enough to be covered by validation, standardized work. Recovery personnel are certified to the recovery process and standardized tools and techniques are used by all recovery personnel.
- **Outliers:** Semiconductor devices that originally passed all manufacturing tests but could be considered at a higher risk of a latent failure compared to other parts in the same population or lot and are therefore more likely to fail in the field.
- **JEDEC:** Formally known as Joint Electronic Device Engineering Council, it is now called the JEDEC Solid State Technology Association. It is an independent semiconductor engineering trade organization and standardization body.
- **ESDA:** Electro Static Discharge Association responsible for developing standards to control Electrical Over Stress or electrically induced physical damage (EIPD).
- **JTAG:** Joint Test Action Group, the common name for what was later standardized as the IEEE 1149.1 Standard Test Access Port and Boundary-Scan Architecture Group.

Appendix B – Definitions & Acronyms: Cyber Security Terminology

- **UNLOCK_ECU_KEY:** A secret key provided by GM along with M1, M2, M3 messages if a security peripheral is required, or as a unique 16- byte value if no security peripheral is required.
- **MASTER_ECU_KEY:** A secret key provided by GM along with M1, M2, M3 messages compliant with the security peripheral memory update protocol.
- **M1, M2, & M3 messages:** Unique values provided by GM to support a security peripheral and associated to a unique ECU_ID value.
- **M4 & M5 messages**: Response values calculated by the individual ECU from the M1,M2 & M3 messages for the purpose of unlocking the ECU for updating software.
- **UDS:** Unified Diagnostic Services
- **Integrated Security Peripheral:**

  An internal semiconductor component with “Secure Hardware Extension” (SHE) / “Hardware Security Module” (HSM) capability.
- **Local Verification Utility Tool:** A method to verify each set of M4 and M5 verification messages returned from the security peripheral as a result of provisioning the MASTER_ECU_KEY and UNLOCK_ECU_KEY.
- **Message Authentication**: A design feature of the cyber security layered protocol.
- **ECU:** Electronic Control Unit, commonly called the electronic module made up of one or more PCB’s
- **Open PGP:** the most widely used encryption standard for all major operating platforms.
- **Authoritative Counter:** An internal counter of messages that have been transmitted by the module (ECU).
- **Active Counter:** An internal counter of messages that have been received by the module (ECU)
- **Security Peripheral Key Slots:**

  The number of general-purpose memory locations (key slots) internal to the SHE/HSM component.
- **Manufacturing Enable Counter:**

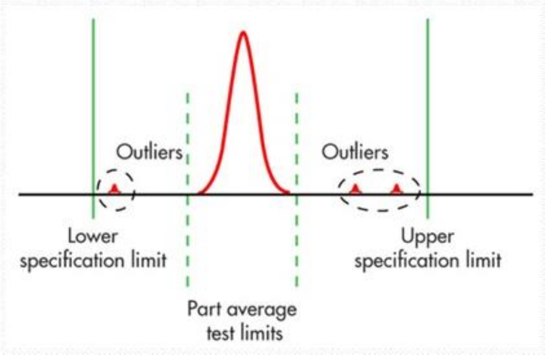
  A digital value (single byte) set to bypass the ECU seed / key feature during product development, ECU manufacturing, and vehicle assembly plant build processes. A value of zero is the locked state and the software cannot be updated.
- **ECU Provisioning Portal:** A GM IT server made available to suppliers for purposes of retrieving data sets to be programmed (provisioned) into each ECU during the ECU manufacturing process.
- **TPISR:** Third Party Information Security Requirements
- **Data set:** The ECU_ID, the UNLOCK_ECU_KEY with associated M1, M2, & M3 messages, and if applicable the MASTER_ECU_KEY with associated M1, M2, & M3 messages. The data set is then associated to a unique ECU_ID.

Appendix B – Definitions & Acronyms

Acronyms, Abbreviations, and Symbols.

AEC	Automotive Electronics Council	AIAG	Automotive Industry Action Group
AOI	Automatic Optical Inspection	APQP	Advanced Product Quality Planning
IATF	**International Automotive Task Force**	CVE	Component Validation Engineer
DRE	Design Release Engineer	ECU:	Electronic Control Unit
GPSC	Global Purchasing & Supply Chain	GQTS	Global Quality Tracking System
PCBA	Printed Circuit Board Assembly	PPAP	Production Part Approval Process
PPQR	Pre-Production Quality Review	PPV	Pre-Production Vehicle
PSW	Part Submission Warrant	SMD	Surface Mounted Device
SMT	Systems Management Team	SMT	Surface Mount Technology
SRV	Supplier Readiness Valve	SQE	Supplier Quality Engineer
SQMS	Supplier Quality Management Systems	VDP	Vehicle Development Plan
PEPU	Per ECU Password Utility	JTAG	Joint Test Action Group
SDC	Supplier Daily Capacity		

Part Average Test limits (as described in section 8.3.4) relative to specification limits showing outliers being removed.



Appendix C – Footnotes

Footnotes for additional explanation of requirements:

[A] Red marked containers or trays for defective material: If E.S.D. containers are used, then red tape along top and sides are acceptable.

[B] Soldering by hand: Solder pots for tinning cable ends is allowed.

[C] The Solder Temperature Impact Index requirements are part of IPC 6010. The STII shall be a minimum 215 for lead free and 240 for advanced PCB designs.

[D] Appearance Approval form (CFG1002) is included in the 4th edition of the AIAG PPAP manual. It is also located in GM Supply Power, Document Library. TALC Approval form (CG2101) is usually included in the eSOR, but can be acquired from the DRE

[E] Part Specific Statements of Requirements are located in GM Supply Power, Document Library, Supplier Quality, Procedures & Forms, Part Specific SOR's 1927 03a folder.

[F] Process Audits are located in GM Supply Power, Document Library, Supplier Quality, Procedures & Forms, Process & Commodity Audits folder.

[G] Internal connectors need a connector-position-assurance activity such as a camera, depth-indicator-line, hole in the connector or a 2nd-push-tool to verify connection is locked properly.

[H] Wooden pallets or dirty card board transport packaging shall not be allowed inside the manufacturing areas. Only cleaned packaging and boxes shall be used for transportation, intermediate storage and shipment. Packaging shall be secured against dirt and foreign material and shall be stored only in appropriate space (no storage outdoors).

[I] ESD shoes should be used instead of shoe strips. ESD arm grounding cables and ESD chairs should be used where workers are sitting. The manufacturing Line shall give an alarm when the worker is not ESD grounded. Gates or alarms shall be used to prevent anyone not wearing ESD equipment from entering the ESD protected area. Also, the ICE-spray used for analysis of overheat shall be ESD compliant.

[J] Random samples at the pointer staking process shall be measured at the start-up and every couple hours thereafter using a variable measurement system. The data shall be plotted on a SPC Run Chart for monitoring variable drift in the equipment.

[K] An "A" versus "B" comparison of the software / hardware shall be completed to guarantee the production manufacturing process is the same as the manufacturing process ~~version~~ used to create the engineering validation samples.

[L] Suppliers can acquire a copy of the latest IPC standards from the official IPC website: IPC.org.

[M] The recommended mechanical control method using a pressure force check for pin presence & position is preferred.

The method of 100% verification using a vision or height measurement system is preferred. All press fit connections shall be verified and documented

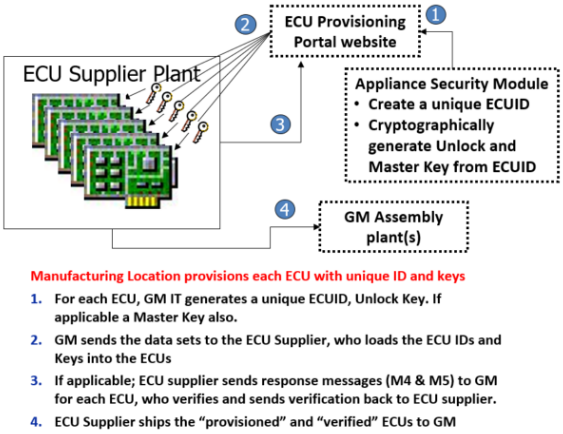
[N] The burst test is a destructive overpressure test where the pressure is increased until it bursts. It must be over a threshold value and also that the burst line is not in the direct contact line of the lid and housing. The two materials should be melted together which gives a stronger connection than in the individual materials.

[O] Appearance of decorative subcomponents of the final assembly (e.g. subcomponents painted and/or laser etched) shall be checked (whether against a master sample or a General Motors' approved color plaque) to ensure there are no color / lighting deviations and/or damage to the subcomponents visible to the customer(s).

[P] In order to get more information on how to obtain a master plaque please contact the GM AAR lead.

[Q] Refer to GM Engineering GB4007 Supplier Controller Provisioning Portal Manual for additional information

[R] ECU_ID, UNLOCK_ECU_KEY & MASTER_ECU_KEY shall be provided in an encrypted format from GM via the ECU Provisioning Portal.



[S] Refer to Third Party Information Security Requirements (TPSIR) for methodology to destroy the M1-M5 values.

Appendix D – Index of sections

Appendix E – Approved deviations to Severity 9 & 10 failure mode detectability

	SOR requirement	Detection capability	Reason for reduced detection capability	Required Controls in place
1	3.11.2 For product on new vehicle launches, involving new hardware and / or new software, the GM 1927 28 exit criteria shall include the following: 3.11.2.1 There shall be no supplier responsible warranty returns at 2 Months in Service (MIS) for 30 production days.	6	Detection of latent defects in tier 2 semiconductor components is impossible until point of failure	Only latent defects in AEC Q100 Integrated Circuits and AEC Q101 Discrete Semiconductors are exempt from this requirement. For these latent defects CG4209 section 8.3.4 applies